# ELISHA MADAKI

**Software and Autonomy Engineer - Embodied Intelligence, Control & Multi-Agent Systems**

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## SUMMARY

Building intelligence that runs on hardware, not just in simulation: control policies embedded in flight firmware, perception models running on the vehicle's own compute, and coordination logic that survives real communication loss. Work spans control, autonomy, and domain-adapted language models for operational tooling. The common thread is embodiment; closing the gap between an algorithm that works in simulation and one that works on hardware with real latency, real noise, and real failure modes. Open to autonomy/software engineering roles and to Research Engineer roles at corporate or institutional labs.

## EXPERTISE

- **Control, Dynamics & Estimation:** Real-Time Nonlinear MPC (acados SQP), SINDy Dynamics Identification, EKF/Kalman Filtering, Geometric & Nonlinear Control, Vector Control.
- **Embedded Flight Software & Security:** Bare-Metal C++17/20, POSIX Real-Time Systems, PX4 Autopilot, ArduPilot Core, MAVLink, uORB, Zero-Trust AEAD Telemetry Encryption (MavlinkCrypt).
- **Perception & Embodied Intelligence:** Onboard Edge Inference, YOLOv8, YOLO11n, SORT/DeepSORT Tracking, Vision-in-the-Loop Setpoint Control, Open-Weight LLM Continued Training/Local Deployment (32K Context), moving toward Deep RL (PPO/SAC).
- **Multi-Agent Systems & Middleware:** ROS 2, Contract-Net Auction Protocols, Decentralized Mesh Networks, Fault-Tolerant Coordination, uXRCE-DDS, rmw_zenoh, Gazebo SITL/HIL, QGroundControl, Mission Planner, Python, Linux/WSL2.

## EXPERIENCE

**Software & Autonomy Engineer** — Briech UAS, Abuja, Nigeria *2025 – Present*

- **Firmware-Embedded NMPC & Dynamic Identification:** Architected and led a team through a real-time C++ NMPC module (*vtol_nmpc_control*) inside PX4 firmware, replacing static PID gains with acados-solved trajectory optimization augmented by SINDy dynamics identification.
- **Decentralized Swarm Coordination:** Built a 5-agent ROS 2 mesh task allocation framework using Contract-Net auctions over YOLOv8-detected task targets, with idempotent re-auctioning for lossy links.
- **Closed-Loop Edge Perception:** Developed an onboard YOLO11n + DeepSORT tracking pipeline, streaming 3D spatial velocity vectors directly into position control loops for autonomous target acquisition.
- **Zero-Trust Telemetry Security:** Architected and led a team implementing *MavlinkCrypt*, a light-footprint AEAD cryptographic layer in flight firmware with automated key zeroing on session teardown.
- **Domain-Adapted LLM for Log Review:** Directed architecture and training-pipeline decisions for an open-weight, 32K-context LLM continued-trained on internal log data and deployed locally via Ollama for automated flight/maintenance log triage.
- **AI-Augmented Fleet GCS:** Led a team modifying and extending QGroundControl to aggregate multi-vehicle telemetry and surface anomalies in the operator UI; separately led a modernization of ArduPilot Mission Planner's workflow (no AI features in that fork). Also built ROS 2/uXRCE-DDS HIL simulation pipelines in Gazebo.

**Software & Autonomy Engineer** — ProForce Air Systems Limited, Ogun, Nigeria *2024 – 2025*

- **Fleet Production Leadership:** Technical lead for a production fleet of 200 autonomous drones built from scratch; end-to-end systems architecture, firmware flashing, setup, configurations and commissioning.
- **Autopilot & Fleet Standardization:** Standardized PX4/ArduPilot flight stacks across multirotor and quadplane platforms.

**Engineer** — Ministry of Aviation & Aerospace Development, Abuja, Nigeria *2024*

- Conducted technical evaluations of autonomous platform flight safety, airworthiness standards, and integrity for regulatory approval.

**Research Engineer, Autopilots** — Air Force Research & Development Centre (AFRDC), Kaduna, Nigeria *2022 – 2023*

- Modified low-level PX4/ArduPilot codebases for experimental military UAS prototypes, integrating specialized sensor payloads and custom control modes.
- Designed Hardware-in-the-Loop test suites evaluating vehicle stability and control authority under nonlinear aerodynamic disturbance.

## EDUCATION

B.Eng., Aerospace Engineering — Air Force Institute of Technology (AFIT), Kaduna, Nigeria | 2018–2023

## SELECTED TECHNICAL CONTRIBUTIONS

Formulation, method, and honest status for the projects referenced above.

**Firmware-Embedded Nonlinear MPC with Data-Driven Dynamics Identification**

*PX4 firmware · C++17 · acados SQP · SINDy · uORB*

**Formulation:** Replace fixed-gain PID with a receding-horizon controller running directly inside PX4's scheduled work loop for VTOL transition, with the dynamics model augmented via SINDy-identified nonlinear terms.

**Method:** Architected the control approach and led a team through implementation: acados-generated SQP solver embedded as a scheduled work item (*px4::ScheduledWorkItem*), running the SINDy-augmented dynamics model on-device at flight-loop frequency. Directed design decisions and reviewed implementation throughout.

**Result:** Running in SITL against the embedded solver; validated for stable trajectory tracking through VTOL transition. Formal benchmarking against the baseline PID controller (tracking error, solve-rate margins) is the next step ahead of flight test.

**Limitation:** SINDy model accuracy has not been characterized outside the trained flight envelope; no formal stability guarantee yet.

**Code:** github.com/MadakiElisha/BriechFirmwarePX/tree/main/src/modules/vtol_nmpc_control

**Decentralized Multi-Agent Task Allocation over Lossy Mesh Networks**

*ROS 2 · ArduPilot SITL · YOLOv8 perception · Contract-Net auctions*

**Formulation:** 5-agent fleet allocating perception-spawned tasks and battery-constrained handoffs without a central coordinator over a live reachability mesh with lossy links. Formulated as a Contract-Net-style auction over YOLOv8-detected task targets.

**Method:** Designed and implemented typed ROS 2 architecture with a *VehicleAdapter* abstraction decoupling coordination logic from the flight controller (ArduPilot/PX4/DJI); watchdog and idempotent re-auctioning recover allocation state after simulated link loss; allocation logic is unit-tested (pytest) independent of flight physics.

**Result:** Coordination validated on ArduPilot SITL's internal physics across simulated link-loss scenarios, with successful re-auctioning recovery. Gazebo's 3D visual rendering is not yet integrated, a lightweight 3D gcs built instead for visualization; affects visualization only, not flight or coordination logic.

**Limitation:** No formal comparison yet against a centralized or market-based allocator baseline from the literature.

**Code:** github.com/MadakiElisha/swarm-projectII

**Closed-Loop Neural Perception for Target Acquisition and Tracking**

*YOLO11n · DeepSORT · Kalman filtering · onboard/edge inference*

**Formulation:** Stream 3D spatial velocity estimates from onboard vision directly into position control loops, without offboard compute, for autonomous target acquisition and tracking.

**Method:** Developed and deployed the edge tracking pipeline: YOLO11n detection frames feed a SORT/DeepSORT Kalman tracker; the resulting velocity estimate is fed directly as a setpoint to the position controller, chosen for its lighter compute footprint at the edge relative to larger YOLO variants.

**Result:** Deployed onboard and flying in closed-loop target-tracking scenarios.

**Limitation:** Performance under low-light or high-occlusion conditions has not been formally characterized; tracking-loss recovery time is not yet quantified.

**Code:** github.com/MadakiElisha/Tracker

**Domain-Adapted, Locally-Deployed Language Model for Flight-Log Review**

*Open-weight base model (Ollama-served) · continued training · local inference · 32K context*

**Formulation:** Automate triage of long, structured flight/maintenance logs where general-purpose models underperform and data sensitivity requires local execution.

**Method:** Continued-trained an open-weight base model on an internally built log data pipeline (32K-token context), served locally via Ollama rather than through a cloud API, keeping operational log data on-premise. ~3-day training run. Gave technical direction, reviewed architecture and data-pipeline decisions, and directed the team executing the run; did not pretrain a model from scratch.

**Result:** Functional local pipeline for automated log triage, in use for flight/maintenance review. A held-out evaluation against human review and against the un-adapted base model is the next step to formally quantify improvement.

**Limitation:** No formal benchmark yet against the un-adapted base model; longest logs are truncated by the 32K context window.

**AI-Augmented Ground Control Station for Fleet Operations**

*QGroundControl (extended) · fleet telemetry aggregation · anomaly detection*

**Formulation:** Stock GCS lacks multi-vehicle fleet supervision, automated tasks, and other custom features.

**Method:** Led a team modifying and extending QGroundControl to aggregate live telemetry across multiple simultaneous vehicles, with anomaly detection surfaced directly in the operator UI rather than requiring manual log review. Separately led a modernization of ArduPilot Mission Planner's interface and workflow (a distinct effort, without AI features).

**Result:** Functional multi-vehicle telemetry aggregation and anomaly-surfacing pipeline. Formal detection-rate benchmarking against labeled fault data is the natural next step.

**Limitation:** Anomaly thresholds are currently hand-tuned rather than learned from labeled fault data.

**Code:** github.com/MadakiElisha/BriechGCS-QGC, github.com/MadakiElisha/BriechGroundControll